Aim :- To study static characteristics of P-N. Junction diodes.

Apparatus :- A given PN- Junction diode, Milliammeter, Voltmeter, D.C. Source, Rheostat connecting wires, etc.

Formula :- (i) Forward static resistance of a diode is given by $R_F = V_F / I_F$.
(ii) Forward dynamic resistance is given by $r_f = \Delta V_F / \Delta I_F$
(iii) Reverse static resistance is given by $R_R = V_R / I_R$.

Procedure :- For Forward Bias
(i) The circuit is assembled as shown in fig 1 in forward biased.
(ii) The Power supply is switched ON.
(iii) The voltage is set at 0V and the current through the diode is recorded with the help of potentiometer the voltage is increased in steps of 0.1V. For each setting of the voltage the corresponding current is noted. The observations are recorded in TABLE - 1. The magni-tude of current will be in the milli ampere range.

Experiment No. 1

Aim :- To study static characteristics of P-N Junc

Apparatus :- A given PN-junction diode, Millimeter, Voltmeter, DC source, Rheostat, co wires, etc.

Formula :- (i) Forward static resistance of a dio given by $R_F = V_F / I_F$

(ii) Forward dynamic resistance is given by $r_F = \Delta V_F / \Delta I_F$.

(iii) Reverse static resistance is given $R_R = V_R / I_R$.

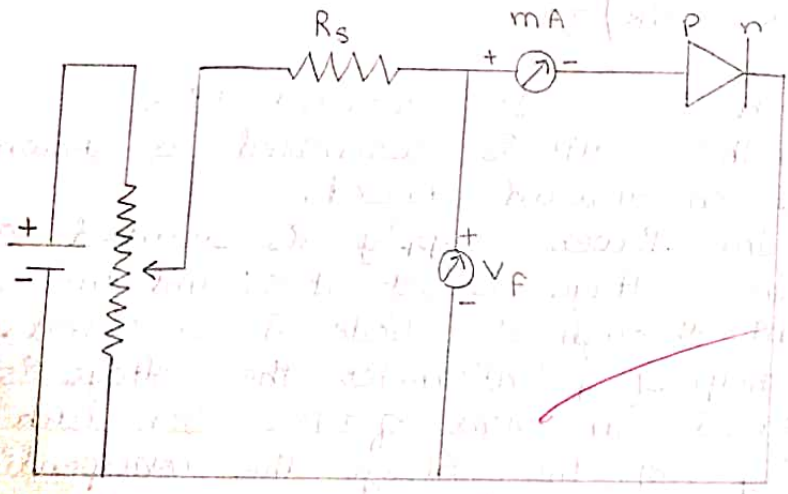


Fig. 1 Circuit connection for forward bias.

For Reverse Bias :-

(i) The circuit is assembled as shown in fig. 2 in Reverse biased.
(ii) The power supply is switched on.
(iii) The voltage is varied in steps of 0.1V.
The current will be in the micro Ampere range the values are recorded in TABLE. I.T. the current reading are noted down. Diode offers high resistance when it is connected in the reverse biased mode.

Result :-

(i) The cut in voltage of diode is found to be 0.2 V
(ii) The forward static resistance $R_F = 2500\,\Omega$
(iii) The reverse static resistance $R_R = 500\,K\,\Omega$
(iv) Forward dynamic resistance $R_F = 0.5 \times 10^3\,\Omega$

Precaution :-
(i) Take reading carefully during forward bias and reverse bias.
(ii) Measure the voltage accurately for reading.

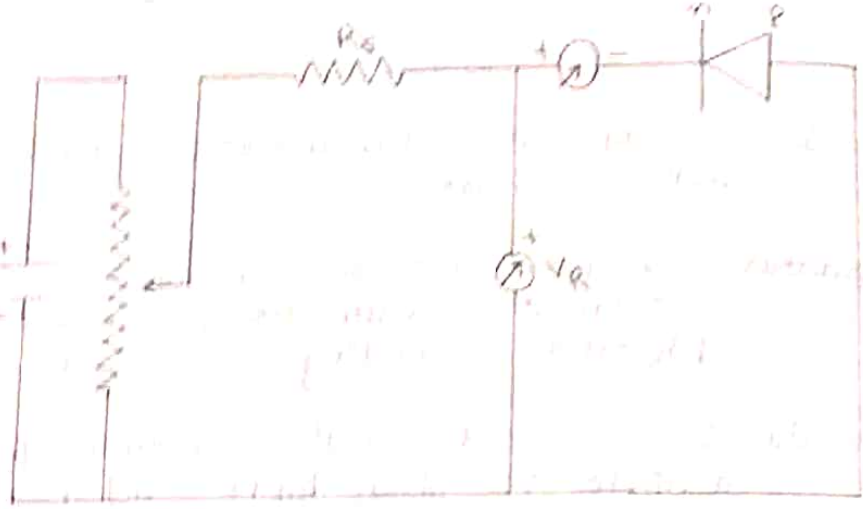


Fig. 2 Circuit connection for Reverse bias

Sr No.	FORWARD BIAS		REVERSE BIAS	
	VF (volts)	IF (mA)	VR (Volts)	IR (μA)
1	0.1	0	1	2
2	0.2	0	2	4
3	0.3	0	3	6
4	0.4	0	4	8
5	0.5	0.2	5	10
6	0.6	0.4	6	12
7	0.7	0.6	7	14
8	0.8	0.6	8	[illegible]
9			9	21
10			10	26

20/9

Calculations :-

$$R_F = \frac{V_F}{I_F} = \frac{0.5}{0.2} = \frac{5}{2\times10^{-3}} = \frac{5000}{2} = 2500\ \Omega$$

(Forward bias Static resis

$$r_F = \frac{\Delta V_F}{\Delta I_F} = \frac{0.5-0.4}{(0.2-0)} = \frac{0.1}{0.2}\ \Omega$$

$$= \frac{0.1}{0.2\times10^{-3}}$$

$$= 0.5\times10^{3}\ \Omega$$ (Forw dynamic

$$R_R = \frac{V_R}{I_R} = \frac{1}{2\times10^{-6}} = \frac{1}{2\times10^{-6}}$$

$$= 0.5\times10^{6}\ \Omega$$ (Reverse

$$= 500\times10^{3}$$ Static

$$= 500\ K\Omega$$

Result :-

i) The cut in voltage of diode is found to

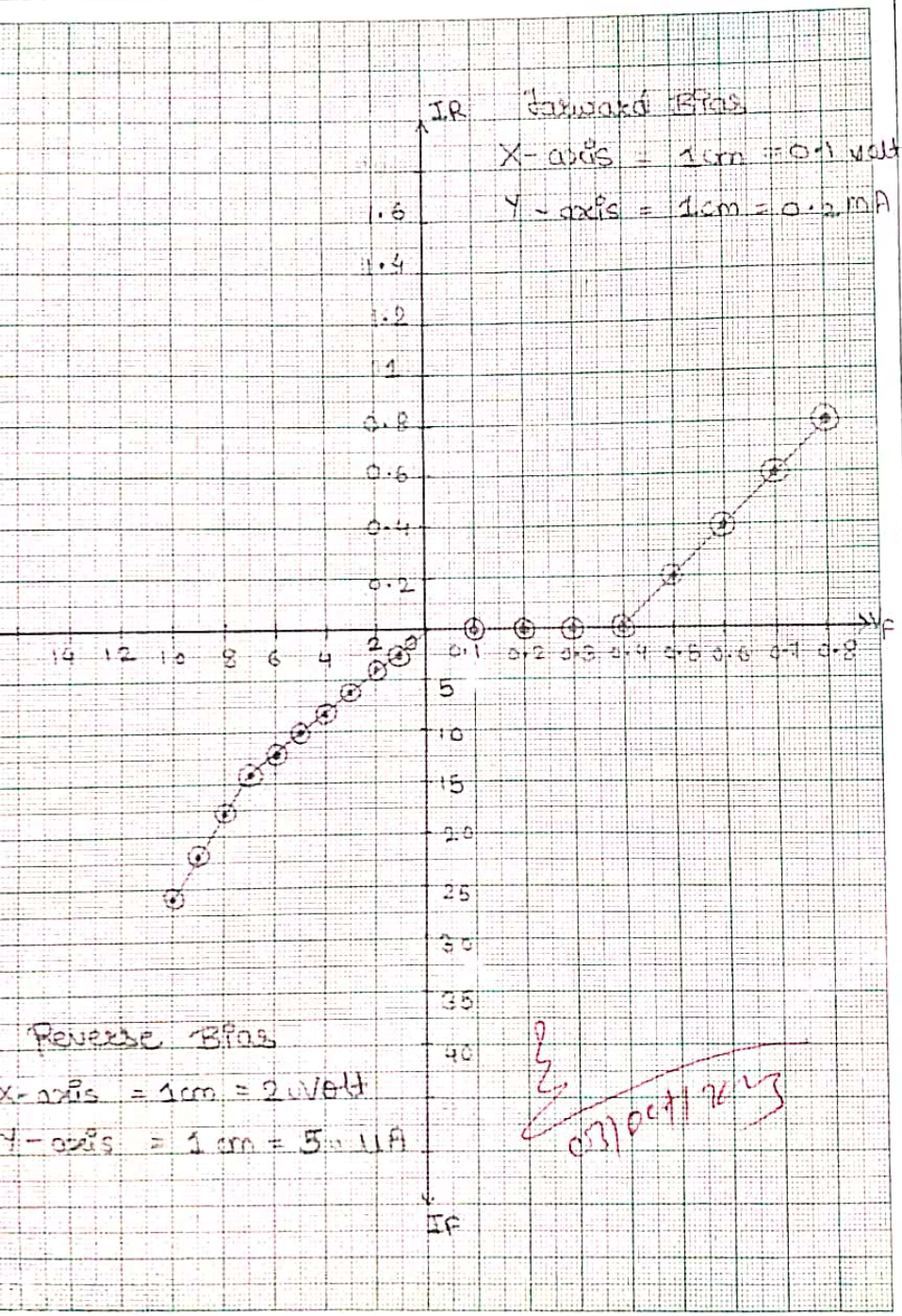
ii) The forward static resistance R_F =

iii) The reverse static resistance R_R =

iv) Forward dynamic resistance r_F = 0.5

03/oct/24

Date .....



Teacher's Signature _____